

The predictive power of autobiographical memory in shaping mental health of young people

Citation

Caitlin Hitchcock, Dou Hong. The predictive power of autobiographical memory in shaping mental health of young people. PROSPERO 2022 CRD42022287786 Available from:
https://www.crd.york.ac.uk/prospERO/display_record.php?ID=CRD42022287786

Review question

This meta-analysis will address two questions investigating whether autobiographical memory prospectively influences mental health young people.

- 1) Does autobiographical memory specificity in young people predict later mental health issues? If so, does the predictive value of memory specificity vary across different symptom clusters?
- 2) Do patient-level characteristics, particularly exposure to stress or trauma, moderate the effect of autobiographical memory specificity on later mental health?

Searches

An international collaborative network will be formed, and members will contribute known datasets. This will be supplemented by formal literature searches in PsycINFO, Web of Science, EMBASE and MEDLINE using the terms autobiographical memory AND adolescen* OR child*

Restrictions will be publication in or after 1986 (as Williams & Broadbent's seminal paper on autobiographical memory specificity was published this year). No language restrictions will be applied to the search, but only papers published in English will be included.

Reference lists of those papers identified will be searched, and papers citing identified articles will also be checked. Furthermore, key authors will be contacted via email to request any unpublished data relevant to the study.

Types of study to be included

Inclusion criteria will be assessment of: i) participants aged 5-20 years; ii) a measure of specificity of autobiographical memory retrieval (e.g., using a cued-recall task or interview); iii) a measure (either self-report or observer-rated) of any symptoms of any mood, stressor-related, anxiety, eating, or psychotic disorder at two time points at least one month apart.

Condition or domain being studied

Symptoms of low mood, anxiety, psychosis, disorder eating and posttraumatic stress.

Participants/population

Inclusion criterion is age of 5-20 years. Exclusion criterion is age over 20 years.

Intervention(s), exposure(s)

The key predictive variable of interest is autobiographical memory specificity, as indexed on the Autobiographical Memory Test (AMT; Williams and Broadbent, 1986). We will calculate the proportion of specific responses for each participant as the number of specific memories retrieved divided by the total number of cues on the AMT.

The secondary predictor of interest is level of episodic detail in reported memories, as per Levine et al.'s original scoring criteria. While this variable may provide a finer index of memory specificity, it is anticipated that fewer datasets will include episodic detail and thus it has been designated as a secondary predictor of interest.

Comparator(s)/control

Not applicable

Main outcome(s)

Symptom score on a self-report measure of a) depression, b) anxiety and c) posttraumatic stress.

For the co-primary outcomes, symptoms will be assessed at least one month following the assessment of autobiographical memory specificity.

Measures of effect

Effect size for the effect of the predictor on the outcome

Additional outcome(s)

Symptom score on a self-report measure of a) disordered eating and b) psychosis. These outcomes have been designated as secondary due to an anticipated low amount of data.

For the co-secondary outcomes, symptoms will be assessed at least one month following the assessment of autobiographical memory specificity.

Measures of effect

Effect size for the effect of the predictor on the outcome

Data extraction (selection and coding)

Two reviewers will independently review title and abstract. Papers deemed ineligible by both reviewers will be excluded. Full text of remaining papers will be independently examined by reviewers. Disagreements will be solved via discussion.

Individual participant data for specified variables will be extracted into a single dataset. Spot checks will ensure data quality.

Variables to be extracted are:

Sex

Age

Ethnicity

Experience of DSM-defined Criterion A trauma

Trauma type

Experience of significant life stress

Family history of mental illness

Prior experience of mental illness

Diagnosis method (clinician/interview/cut-off on self report measure/report on medical records)

Current psychotropic medication use

Completion of psychological treatment use during study period

Number of weeks between baseline and follow-up assessment

Name of self-report symptom measure

Baseline score on self-report measure of symptoms

Name of observer-report symptom measure

Baseline observer rating of symptoms (continuous)

Baseline observer rating of symptoms (diagnostic status)

Baseline total number of specific memories on AMT

Total number of word cues on AMT

Baseline number of specific memories to positive cues on AMT

Baseline number of specific memories to negative cues on AMT

Baseline number of specific memories to neutral cues on AMT

Number of positive word cues on AMT

Number of negative word cues on AMT

Number of neutral word cues on AMT

Baseline level of episodic detail for positive memories

Baseline level of episodic detail for negative memories

Baseline level of episodic detail for neutral memories

Follow-up score on self-report measure of symptoms

Follow-up observer rating of symptoms (continuous)

Follow-up observer rating of symptoms (diagnostic status)

Follow-up total number of specific memories on AMT

Follow-up number of specific memories to positive cues on AMT

Follow-up number of specific memories to negative cues on AMT

Follow-up number of specific memories to neutral cues on AMT

Follow-up level of episodic detail for positive memories

Follow-up level of episodic detail for negative memories

Follow-up level of episodic detail for neutral memories

Any potential cognitive covariates (e.g., rumination, IQ)

Reasons for missing data

Risk of bias (quality) assessment

Risk of bias for included studies will be evaluated independently by two reviewers using the revised Cochrane Risk of Bias Tool (RoB 2) to assess study quality and risk of bias in:

1. Bias arising from the randomization process
2. Bias due to deviations from intended interventions
3. Bias due to missing outcome data
4. Bias in measurement of the outcome
5. Bias in selection of the reported result
6. Overall bias

Using the tool, each study will be rated as of high-, low-, or unclear risk. Any disagreements between reviewers will be resolved via discussion. If there is significant differences in the quality of studies, this will be accounted for in follow-up analysis.

Strategy for data synthesis

Individual patient data (IPD) will be used.

Between-study heterogeneity will be indexed using the tau statistic. If IPD are available for more than seven studies, prediction intervals will be used (Partlett & Riley, 2017).

A one-stage IPD meta-analysis approach will be employed. Analysis of co-primary and co-secondary outcomes will be completed using a random effects model employing restricted maximum likelihood estimation with the Hartung-Knapp method, in which memory specificity at baseline predicts symptom score at follow-up, controlling for baseline symptom score.

Analysis of subgroups or subsets

The below analyses will be completed if a suitable amount of data is obtained.

Length of follow-up: studies including a follow-up assessment between one and three months after baseline will be grouped to form a short-term follow-up, and any later assessment points will be grouped per six month period (i.e., 6 months, 12 months). The primary analysis will be repeated to evaluate of the strength of the predictive effect of memory specificity on prognosis in the short-term (1-3 months), and 6, 12, 18, and 24 months later.

Moderator analysis: We will also explore if the predictive effect of memory specificity on mental health outcomes is

moderated by

- Participant exposure to trauma
- Participant exposure to significant life stress
- Participant prior experience/diagnosis of mental illness
- Participant family history of mental illness
- Participant age
- Emotional valence (positive, negative or neutral) of the word cues used on the AMT
- Sex assigned at birth

For these analyses, a one-stage model predicting symptoms at follow-up will estimate the interaction between baseline memory specificity and each moderator listed above, applying a baseline symptom adjustment.

Contact details for further information

Caitlin Hitchcock

caitlin.hitchcock@unimelb.edu.au

Organisational affiliation of the review

Melbourne School of Psychological Sciences, University of Melbourne

MRC Cognition and Brain Sciences Unit, University of Cambridge

Review team members and their organisational affiliations

Dr Caitlin Hitchcock. MRC Cognition and Brain Sciences Unit

Ms Dou Hong. University of Melbourne

Type and method of review

Individual patient data (IPD) meta-analysis, Prognostic, Systematic review

Anticipated or actual start date

03 January 2022

Anticipated completion date

28 December 2023

Funding sources/sponsors

Australian Research Council

Grant number(s)

State the funder, grant or award number and the date of award

Discovery Early Career Researcher Award, DE200100043

Conflicts of interest

Language

English

Country

Australia, England

Stage of review

Review Ongoing

Subject index terms status

Subject indexing assigned by CRD

Subject index terms

Adolescent; Humans; Memory, Episodic; Mental Health; Mental Recall

Date of registration in PROSPERO

07 January 2022

Date of first submission

19 December 2021

Stage of review at time of this submission

The review has not started

Stage	Started	Completed
Preliminary searches	No	No
Piloting of the study selection process	No	No
Formal screening of search results against eligibility criteria	No	No
Data extraction	No	No
Risk of bias (quality) assessment	No	No
Data analysis	No	No

The record owner confirms that the information they have supplied for this submission is accurate and complete and they understand that deliberate provision of inaccurate information or omission of data may be construed as scientific misconduct.

The record owner confirms that they will update the status of the review when it is completed and will add publication details in due course.

Versions

07 January 2022